60-Day COBOL Interview Study Plan

1-2 hours per day | Interview-focused | COBOL + SQL + batch + JCL + production troubleshooting

Daily target	30 min theory + 45 min coding/practice + 15-30 min interview questions
Main goal	Be able to explain COBOL fundamentals and reason through real production batch problems confidently.
Best approach	Type and run code yourself. Predict the output before compiling. Explain each concept aloud as if the interviewer asked you.

How to use this checklist

Mark each box only after you can both explain the concept and use it in a small program or scenario. If a topic is weak, carry it forward into the next review day instead of skipping it.

Days 1-7 - COBOL Core Refresh

Day	Study checklist	Practical / interview task
1	☐ COBOL program structure ☐ IDENTIFICATION DIVISION ☐ ENVIRONMENT DIVISION ☐ DATA DIVISION ☐ PROCEDURE DIVISION	☐ Write Hello World + simple variables ☐ Explain all 4 divisions aloud
2	☐ PIC X ☐ PIC 9 ☐ PIC S9 ☐ V implied decimal ☐ VALUE clause	☐ Create an employee record with name, ID and salary ☐ Explain the valid range for each field
3	☐ MOVE ☐ INITIALIZE ☐ Numeric vs alphanumeric moves ☐ Padding and truncation	☐ Test MOVE between different field sizes ☐ Predict each result before running the program
4	☐ ADD ☐ SUBTRACT ☐ MULTIPLY ☐ DIVIDE ☐ COMPUTE	☐ Build salary + allowance + tax calculation ☐ Practice ROUNDED and ON SIZE ERROR basics
5	☐ IF / ELSE ☐ Nested IF ☐ Condition names (88-level preview) ☐ Comparison operators	☐ Create a salary classification program ☐ Explain when nested IF becomes hard to maintain
6	☐ EVALUATE ☐ EVALUATE TRUE ☐ Multiple WHEN conditions	☐ Rewrite Day 5 nested IF using EVALUATE ☐ Compare readability with IF / ELSE
7	☐ PERFORM ☐ PERFORM UNTIL ☐ PERFORM VARYING ☐ Paragraph execution	☐ Create loop processing 10 employees ☐ Review Days 1-6 with 10 interview questions

Days 8-14 - Data Structures and Core Features

Day	Study checklist	Practical / interview task
8	☐ Level 01 ☐ Level 05 / 10 / 15 ☐ Level 77 ☐ Group vs elementary items	☐ Design a customer record hierarchy ☐ Explain group movement
9	☐ Level 88 condition names ☐ SET condition TO TRUE ☐ Readable status checks	☐ Create ACTIVE / INACTIVE / SUSPENDED status flags ☐ Replace literal comparisons with 88-level conditions
10	☐ REDEFINES ☐ Shared storage concept ☐ Common use cases	☐ Split YYYYMMDD into year / month / day ☐ Explain why REDEFINES does not copy data
11	☐ OCCURS ☐ Tables / arrays ☐ Subscripts	☐ Store 20 employees in a table ☐ Loop through all entries
12	☐ INDEXED BY ☐ SET index ☐ Index vs subscript	☐ Traverse OCCURS table using an index ☐ Explain index vs numeric subscript
13	☐ SEARCH ☐ SEARCH ALL ☐ Sorted table requirement	☐ Implement linear search ☐ Implement binary search and explain prerequisites
14	☐ COPYBOOK ☐ COPY ☐ FILLER ☐ RENAMES basics	☐ Build reusable employee copybook ☐ Review REDEFINES / OCCURS / SEARCH / copybooks

Days 15-21 - File Processing

Day	Study checklist	Practical / interview task
15	☐ SELECT ☐ ASSIGN ☐ FILE-CONTROL ☐ FD record definition	☐ Define an employee input file ☐ Explain relation between SELECT and FD
16	☐ OPEN INPUT ☐ OPEN OUTPUT ☐ OPEN EXTEND ☐ CLOSE	☐ Read an input file and create output ☐ Test missing-file behavior
17	☐ READ ☐ AT END ☐ EOF flag pattern	☐ Build standard read-until-EOF loop ☐ Explain why EOF must be controlled explicitly
18	☐ WRITE ☐ REWRITE ☐ DELETE ☐ Record update concepts	☐ Modify records and write output ☐ Explain requirements for REWRITE
19	☐ FILE STATUS ☐ Success / not found / duplicate-key concepts	☐ Add FILE STATUS checks ☐ Create a simple error message routine
20	☐ Sequential files ☐ Line sequential files ☐ Indexed files	☐ Compare access patterns ☐ Relate line sequential processing to GnuCOBOL / NetCOBOL
21	☐ Relative file basics ☐ File organization review ☐ Error-record design	☐ Mini project: input -> validation -> valid.dat / error.dat ☐ Explain EOF and file-error handling end-to-end

Mini-project target: employee.dat -> COBOL validation -> valid.dat + error.dat. Add EOF handling and FILE STATUS checks.

Days 22-30 - SQL and COBOL

Day	Study checklist	Practical / interview task
22	☐ SELECT ☐ INSERT ☐ UPDATE ☐ DELETE	☐ Write common employee SQL statements ☐ Practice WHERE conditions carefully
23	☐ INNER JOIN ☐ LEFT JOIN ☐ GROUP BY ☐ HAVING	☐ Write 5 interview-style SQL queries ☐ Explain WHERE vs HAVING
24	☐ Primary key ☐ Foreign key ☐ Indexes ☐ Uniqueness	☐ Explain when an index helps ☐ Explain index write-cost trade-off
25	☐ Transactions ☐ COMMIT ☐ ROLLBACK ☐ Atomicity concept	☐ Explain partial batch failure ☐ Design safe commit points
26	☐ Embedded SQL syntax ☐ Host variables ☐ EXEC SQL / END-EXEC	☐ Write COBOL SELECT INTO example ☐ Explain colon-prefixed host variables
27	☐ SQLCODE ☐ SQLSTATE ☐ No-row and error handling	☐ Add SQL error handling ☐ Map common outcomes to program actions
28	☐ Cursor concept ☐ DECLARE CURSOR ☐ OPEN	☐ Create cursor for employee rows ☐ Explain why a cursor is used
29	☐ FETCH ☐ Cursor loop ☐ CLOSE	☐ Process multiple DB rows ☐ Handle end-of-cursor condition
30	☐ DB performance ☐ Indexes ☐ Full table scans ☐ Excessive DB calls ☐ Commit frequency	☐ Answer: batch processing 5M rows became slow - what do you investigate? ☐ Practice a structured performance-troubleshooting answer

Day 30 target: explain SELECT INTO vs cursor, commit strategy, and how you would investigate a batch job that suddenly became slow.

Days 31-38 - JCL and Batch Processing

Day	Study checklist	Practical / interview task
31	☐ JOB ☐ EXEC ☐ DD ☐ Job -> Step -> Program model	☐ Explain each concept aloud ☐ Draw a simple two-step job
32	☐ DSN / dataset concepts ☐ SYSOUT ☐ Input and output DDs	☐ Read a sample JCL ☐ Identify each resource used by the program
33	☐ DISP concepts ☐ NEW / OLD / SHR ☐ Dataset lifecycle	☐ Explain dataset sharing and creation concepts ☐ Relate mainframe terminology to files you used on Windows
34	☐ Condition codes ☐ Return codes ☐ Step success / warning / failure	☐ Explain how later steps react to prior results ☐ Create a return-code decision example
35	☐ Multi-step jobs ☐ Dependencies ☐ Batch flow	☐ Draw a four-step nightly batch workflow ☐ Identify upstream and downstream effects
36	☐ Restart concepts ☐ Checkpoint thinking ☐ Safe rerun	☐ Answer: STEP03 fails - how do you restart safely? ☐ Discuss duplicate-processing risk
37	☐ GDG basics ☐ Temporary datasets ☐ Generation concepts	☐ Explain why batch systems keep file generations ☐ Describe recovery benefits
38	☐ JES concept ☐ SDSF concept ☐ z/OS terminology overview	☐ Explain your NetCOBOL/Windows background accurately ☐ Map familiar batch concepts to mainframe terms

Days 39-46 - Production Debugging

Day	Study checklist	Practical / interview task
39	☐ Failed batch job ☐ Failed-step isolation ☐ Return code / logs / SQLCODE / file status	☐ Walk through investigation in order ☐ State impact before rerunning
40	☐ Invalid input ☐ Bad-record isolation ☐ Reject handling	☐ Trace one malformed record ☐ Explain correction and reprocessing
41	☐ Duplicate data ☐ Restart logic ☐ DB keys ☐ Commit boundaries	☐ List likely causes ☐ Explain how to prevent duplicate reprocessing
42	☐ Slow batch ☐ Input volume ☐ SQL / index ☐ CPU / RAM / disk I/O	☐ Build a troubleshooting checklist ☐ Separate application vs infrastructure causes
43	☐ Database failure ☐ COMMIT ☐ ROLLBACK ☐ Restart point	☐ Explain partial-processing recovery ☐ Design verification after rerun
44	☐ File / disk problem ☐ Missing files ☐ Permissions ☐ Capacity	☐ Use FILE STATUS thinking ☐ Explain operational checks
45	☐ Job dependency failure ☐ Scheduler ☐ Upstream / downstream impact	☐ Decide whether downstream jobs can continue ☐ Describe safe recovery order
46	☐ Incident communication ☐ Root cause ☐ Fix ☐ Prevention	☐ Tell one incident using Symptom -> Evidence -> Investigation -> Root cause -> Fix -> Prevention ☐ Keep answer under 3 minutes

Use this answer structure: Symptom -> Evidence -> Investigation -> Root cause -> Fix -> Prevention.

Days 47-52 - Advanced COBOL

Day	Study checklist	Practical / interview task
47	☐ CALL ☐ Called program ☐ Subprograms	☐ Write Program A calling Program B ☐ Explain modular COBOL
48	☐ LINKAGE SECTION ☐ PROCEDURE DIVISION USING ☐ Parameter layout	☐ Pass employee data between programs ☐ Explain caller / callee agreement
49	☐ BY REFERENCE ☐ BY CONTENT	☐ Demonstrate whether callee changes affect caller ☐ Explain use cases
50	☐ Static CALL ☐ Dynamic CALL ☐ Deployment trade-offs	☐ Explain high-level differences ☐ Relate to maintainability
51	☐ COMP ☐ COMP-3 ☐ Binary vs packed decimal ☐ Storage concepts	☐ Explain why COMP-3 is common for money ☐ Practice simple storage-size questions
52	☐ Reference modification ☐ STRING ☐ UNSTRING ☐ INSPECT	☐ Parse a delimited employee record ☐ Count / replace characters using INSPECT

Days 53-55 - Mainframe Concepts

Day	Study checklist	Practical / interview task
53	☐ VSAM purpose ☐ KSDS ☐ ESDS ☐ RRDS ☐ Key-based access	☐ Focus on KSDS ☐ Compare VSAM and sequential files
54	☐ CICS purpose ☐ Online transaction processing ☐ COMMAREA ☐ BMS ☐ Transaction concept	☐ Explain batch vs online COBOL ☐ Draw User -> CICS -> COBOL -> DB / VSAM
55	☐ Enterprise architecture ☐ COBOL + Java / C# integration ☐ Batch + APIs ☐ File transfer ☐ Scheduling	☐ Answer: how can a legacy COBOL system integrate with modern services? ☐ Use examples from your own technology background

Days 56-60 - Interview Mode

Day	Study checklist	Practical / interview task
56	☐ Four COBOL divisions ☐ PIC clauses ☐ COMP-3 ☐ REDEFINES ☐ OCCURS ☐ SEARCH / SEARCH ALL ☐ Index vs subscript ☐ 88-level ☐ Copybook ☐ CALL / LINKAGE	☐ Answer 15 COBOL questions without notes ☐ Keep each answer clear and concise
57	☐ SELECT INTO vs cursor ☐ SQLCODE ☐ COMMIT / ROLLBACK ☐ Indexes ☐ JOINs ☐ JOB / EXEC / DD ☐ Return codes ☐ Restart	☐ Answer: batch updates 500,000 rows and crashes after 300,000 - what next? ☐ Explain data consistency and rerun safety
58	☐ Job failure ☐ Missing file ☐ SQL error ☐ Duplicate records ☐ Slow batch ☐ Disk full ☐ Invalid input ☐ DB connection failure ☐ Dependency failure ☐ Wrong output totals	☐ Solve 10 production scenarios ☐ Use Check -> Isolate -> Impact -> Recover -> Verify -> Prevent
59	☐ Tell me about yourself ☐ COBOL experience story ☐ Production incident ☐ Performance problem ☐ Legacy maintenance ☐ Java / C# integration ☐ Why change jobs?	☐ Prepare 2-3 minute career introduction ☐ Prepare 5 STAR stories
60	☐ Mock interview only ☐ No new material ☐ Identify final weak areas	☐ 30 min COBOL + 20 min SQL + 20 min JCL/batch + 20 min troubleshooting + 20 min career ☐ Write down only your 5 weakest areas for final review

Fedora Practice Setup

For daily COBOL practice on Fedora, use GnuCOBOL. It is ideal for the language features in this plan: PIC clauses, MOVE, arithmetic, IF/EVALUATE, PERFORM, OCCURS, SEARCH, files, copybooks, CALL/LINKAGE, COMP-3, STRING/UNSTRING, and more.

Recommended tools

Tool	Purpose	Fedora note
GnuCOBOL	Compiler and runtime	sudo dnf install gnucobol
Git	Track all 60 days	sudo dnf install git
GDB	Debug native executable when useful	sudo dnf install gdb
Make	One-command builds	sudo dnf install make
PostgreSQL	SQL practice	Install when you reach Days 22-30

Compile and run

cobc -x -free hello.cob -o hello
./hello

Suggested folder structure

~/cobol-study/day01 ... day60, plus copybooks/, files/, sql/, and projects/. Keep one README with short notes on what you learned each day.

For JCL, VSAM and CICS

Do not spend most of your two months trying to emulate a full IBM mainframe locally. For interview preparation, prioritize concepts, reading sample JCL, production troubleshooting, restart logic, and accurate explanations of what you used in NetCOBOL / Windows environments.

Final Day 60 target

You should be able to explain a complete flow: Scheduler -> Job/JCL -> COBOL program -> input files/database -> SQL cursor/commit -> output/report, then troubleshoot what happens if a middle step fails after partial processing.